Narrated Accessibility Walk

Davis Square to Assembly Square, Somerville, MA: College Ave, Powderhouse Square, Broadway, Winter Hill, Mystic Ave and under I-93

Pilot report 7 | Recorded November 1, 2026

Part of an independent series of narrated walks that check whether people with disabilities can reach clinical research sites.
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Summary

Unlike pilot walks 1 to 6, this one did not go to a research site. It was an everyday trip on foot, from Davis Square to the shops at Assembly Square, along College Ave, Broadway and Mystic Ave and under I-93. It tests whether page 1 of the Route to Research checklist works for any destination. It does: the walk met the same kinds of barriers as the routes to hospitals.

- **24 items recorded:** 3 rated high, 9 medium or mixed, 5 low or notes, and 7 positive features.
- **Crossings for blind and low-vision people were the clearest contrast.** At Broadway and Willow Ave, a signal for blind pedestrians could be heard and had a braille sign (A9). Near Assembly Square, a push button had no speaker at all (C9), and at Winter Hill the signal was hard to hear over traffic (C1).
- **A crosswalk with no way out.** Near I-93, a crosswalk ends at gravel and broken ground with no curb ramp (C7). A wheelchair user could cross the street and then be stuck.
- **A hazard in the walking area.** On Broadway, just past the bridge over the rail line, the base of a removed pole has four bolts sticking up, in an area with no street light noticed (B2).
- **Rest.** The first benches were about a mile into the walk (A5); other stretches had none (C4).
- **Drivers.** At Temple St and Mystic Ave, a driver blocked the crosswalk during the walk signal (C5).
- **New work helps.** New curb ramps with warning strips under I-93 (C6), a very wide new path to Assembly Square (C8) and a generous countdown at one crossing (C10).

Each item is a possible barrier seen by one walker. None is a legal compliance finding. Places marked "to be confirmed" will be checked on the next walk before any request is filed.

Key findings in photos



Figure 1. Broadway, just east of the bridge over the rail line: the base of a removed pole with four anchor bolts sticking up in the walking area (B2).



Figure 2. The crossing near I-93 on the way to Assembly Square: the push button stands in gravel and broken ground where the crosswalk ends, with no curb ramp (C7). The hand is the walker's.



Figure 3. A push button near Assembly Square with no speaker: it gave no sound at all (C9).



Figure 4. Broadway at Willow Ave: a signal for blind pedestrians, with a sign reading "Push button for accessible signal". The walk signal could be heard (A9, positive).



Figure 5. Mystic Ave under I-93: a new curb ramp with a warning strip (C6, positive). The walk signal here felt short.



Figure 6. The approach to Assembly Square: a new, very wide paved path (C8, positive).

Walk details

Item	Details
Date	November 1, 2026 (Sunday), afternoon, sunny and dry
Recordings	Three narrated phone videos, about 39 minutes in all (18 min 31 s, 2 min 25 s and 18 min 28 s)
Route	Davis Square → College Ave → Powderhouse Square → Broadway → Ball Square (video 1); Broadway bridge over the rail line, east of Ball Square (video 2); Broadway on Winter Hill → Temple St → Mystic Ave → under I-93 → Assembly Square (video 3). Short stretches between the videos were not recorded.
Narrator	Uri Kartoun, PhD (DBbun LLC); assessed the route from the point of view of wheelchair, walker and cane users, blind and low-vision people, and people who tire easily. The walker does not use a mobility aid.
Why this route	An everyday destination rather than a research site: a test of whether page 1 of the checklist works for any trip. Assembly Square has a rail station and large stores; the walk reached it on foot from the north and west, across busy roads and highway ramps.
Planning	Not planned from the open-data screen used on earlier walks; this was a walk the author was taking anyway, narrated as it happened.

What travelers would meet

What each group of travelers would likely meet, based on what the walker observed. These are not test results: no traveler with these disabilities made the trip, so each cell notes what was seen and, where it matters, what remains uncertain.

Traveler	Along the sidewalks	At crossings	Main items
Wheelchair (manual or power)	Seen: cracks and lifted slabs on Broadway (A7); a broken edge at a closed gas station driveway (B4); a very wide new path near Assembly Square (C8, positive). Uncertain: the slope of a church ramp (A3) and a side-street curb ramp (C3), not measured.	Seen: a crosswalk that ends at gravel with no curb ramp (C7); a crosswalk blocked by a driver (C5); new curb ramps under I-93 (C6, positive).	C7, A7, C5
Walker, rollator or cane; people who tire easily	Seen: the first benches about a mile in (A5); none on Broadway toward Mystic Ave (C4); several on Broadway in the 720s–740s (A8, positive).	Seen: a generous countdown at one crossing (C10, positive); a walk signal under I-93 that felt short (C6). Uncertain: actual signal times, not measured.	A5, C4, C6
Blind or low vision	Seen: a pole base with bolts sticking up in the walking area (B2); an e-bike ridden through the Davis plaza (A1); overgrowth (B3).	Seen: an audible signal with braille at Willow Ave (A9, positive); a signal hard to hear over traffic (C1); a push button with no speaker (C9); a store entrance with no crosswalk lines (C11).	C9, B2, C1
Deaf or hard of hearing	No barriers specific to this group were noted.	Seen: visual walk signals and countdowns at the signalized crossings.	–

Findings

Rating: High = can stop someone from getting through or cause a fall; Medium = makes the trip much harder; Low = a nuisance; Positive = something that works and should be kept or copied. Conf. (confidence): H = seen in the video frames; M = described in the narration only, or seen once and not measured. Sizes and counts are estimates unless stated. Items are numbered by section (A, B, C); A4 and B5 are not shown because they concern buildings left out by protocol.

A. Davis Square, College Ave and Powderhouse Square, along Broadway to Ball Square

ID	Place	What was seen	Most affected	Rating	Conf.	Suggested fix
A1	Davis Square plaza	An e-bike was ridden through the pedestrian plaza; bumpy brick in places.	Blind people, slow walkers	Medium	M	Signs or design that keep bikes slow in the plaza
A2	College Ave near the Davis station entrance	A fresh asphalt patch at the curb ramp and crosswalk, and a smooth new sidewalk after it.	–	Positive	H	–
A3	A church on College Ave	Front steps with handrails; a separate ramp with a noticeable slope (not measured).	Manual wheelchair users	Low (note)	M	Measure the slope on a later visit
A5	Powderhouse Square park	The first benches (three) since Davis Square, about a mile into the walk.	People who need to rest	Medium	H	Seats along College Ave
A6	Crossing at Powderhouse Square	Wide crosswalks, but nothing controlled traffic coming from the left; it felt unsafe.	Slow walkers, wheelchair users	Medium	M	Review traffic control at this crossing
A7	Broadway, College Ave to Liberty Ave	Several cracks and lifted slabs, about 2 inches by eye; one large crack before Liberty Ave.	Wheelchair users, cane users	Medium	M	Sidewalk repair (addresses to be logged on the re-check walk)
A8	Broadway, 740s to 720s	Several benches along this stretch.	–	Positive	M	–
A9	Broadway at Willow Ave	Signals for blind pedestrians, with signs and braille; the walk signal could be heard. Some people crossed on red.	Blind and low-vision people	Positive	H	A model for the crossings in C1 and C9

ID	Place	What was seen	Most affected	Rating	Conf.	Suggested fix
A10	Ball Square (Green Line station)	Clear signs to the station.	–	Positive	M	–

B. Broadway bridge over the rail line, east of Ball Square

ID	Place	What was seen	Most affected	Rating	Conf.	Suggested fix
B1	Broadway bridge over the rail line	Wide sidewalk, separate bike lane and a high fence; the metal joint plate looked flush.	–	Positive	H	–
B2	Broadway, just past the east end of the bridge	The base of a removed pole with four anchor bolts sticking up in the walking area. Hard to see, especially at night; no street light was noticed here.	Blind and low-vision people, cane users, anyone at night	**High**	H	Remove the bolts or cover the base; check the lighting
B3	Same stretch	Dry weeds along the fence, a bush growing over the sidewalk, a leaning sign post.	Wheelchair users, blind people	Low	H	Trim; straighten the post
B4	Driveway of a closed gas station	A wide curb cut across the sidewalk, with broken pavement and crushed traffic cones at the edge.	Wheelchair users, slow walkers	Low–Medium	H	Repair the edge; remove the debris

C. Winter Hill, Temple St and Mystic Ave, under I-93, to Assembly Square

ID	Place	What was seen	Most affected	Rating	Conf.	Suggested fix
C1	Broadway crossing, Winter Hill	The signal for blind pedestrians said "wait", but it was hard to hear over traffic.	Blind and low-vision people	Medium	M	Check the volume setting
C2	Broadway	Construction beside the sidewalk; the crossing was still possible.	–	Low	M	Keep a step-free, signed path during the works
C3	A side street off Broadway	A curb ramp somewhat steep.	Manual wheelchair users	Low	M	Measure on a later visit
C4	Broadway toward Mystic Ave	No benches noticed on this stretch.	People who need to rest	Low	M	Seats where possible
C5	Temple St at Mystic Ave	A driver blocked the crosswalk during the walk signal; cones on the sidewalk edge.	Everyone crossing; worst for wheelchair users and slow walkers	Medium	M	Enforcement; "do not block the crosswalk" marking
C6	Mystic Ave under I-93	Detour signs on the sidewalk; new curb ramps with warning strips. The walk signal felt short (to be timed).	Slow walkers	Mixed	H	Time the signal on the re-check walk
C7	Crossing near I-93 on the way to Assembly Square (exact place to be confirmed)	The crosswalk ends at gravel and broken ground, with no curb ramp: there is no way for a wheelchair to leave the street. Construction nearby.	Wheelchair users, walker users, strollers	**High**	M	A curb ramp or a temporary step-free path
C8	Approach to Assembly Square	A new, very wide paved path.	–	Positive	H	–
C9	A busy crossing near Assembly Square (exact place to be confirmed)	A push button with no speaker: it gave no sound at all, unlike the signals for blind pedestrians elsewhere on the walk.	Blind and low-vision people	**High**	H	A signal that beeps or talks, as at Willow Ave (A9)
C10	Same area	A generous countdown: 24 seconds were left when the walker reached the other side.	Slow walkers	Positive	M	–

ID	Place	What was seen	Most affected	Rating	Conf.	Suggested fix
C11	Store parking entrance at Assembly Square	Cars in and out at a bottleneck, no crosswalk lines, and broken asphalt right after the warning strip.	Blind people, wheelchair users	Medium	H	Mark the crossing; repair the asphalt

What this walk shows for research visits

Many research participants travel to sites along streets like these. A study team that sends a participant only "the address" leaves them to find out on the day about a crossing with no sound or a crosswalk with no curb ramp. The same checklist items apply wherever the visit is: crossings that beep or talk, a curb ramp at both ends of every crosswalk, places to sit, and step-free, signed paths around construction. The walk also shows how one city can have both: a model crossing for blind pedestrians on Broadway (A9) and, a mile or two away, crossings with a weak or missing sound (C1, C9).

Evidence for key findings

Where each key finding can be checked in the recordings. The recordings are kept by the author and are not published, because they show people and license plates; excerpts with faces and plates blurred are available on request to the City.

ID	Finding	Where to find it (video, time)
B2	Pole base with bolts in the walking area	Video 2, 1:00–1:10. Figure 1.
C7	Crosswalk ends at gravel, no curb ramp	Video 3, 12:30–12:53. Figure 2.
C9	Push button with no speaker	Video 3, 16:10–16:38. Figure 3.
C1	Signal hard to hear over traffic	Video 3, 0:36–0:50. Figure 7.
A9	Audible signal with braille (positive)	Video 1, 14:13–15:21. Figure 4.
C5	Driver blocking the crosswalk	Video 3, 7:14–7:31.
A7	Cracks and lifted slabs on Broadway	Video 1, 10:34–11:42.

Questions for the City, the state and property owners

- City of Somerville: could the pole base on Broadway east of the rail bridge (B2) be removed or covered, and the lighting there checked?
- City of Somerville: could the volume of the signal for blind pedestrians at the Broadway crossing on Winter Hill (C1) be checked?
- City of Somerville or the state highway department, whichever owns the crossing: could the crosswalk near I-93 that ends at gravel (C7) get a curb ramp or a temporary step-free path, and could the push button near Assembly Square (C9) be replaced with one that beeps or talks, like the one at Broadway and Willow Ave (A9)?
- City of Somerville: could the crossing at Powderhouse Square (A6) and the sidewalk on Broadway between College Ave and Liberty Ave (A7) be reviewed?
- Property owners at Assembly Square: could the store parking entrance (C11) get crosswalk lines and an asphalt repair?
- Study teams anywhere: do directions for participants with disabilities mention crossings without sound and the step-free way around construction?

Next steps

- Re-check walk (November 16–28, 2026): confirm the exact places of C7 and C9, time the walk signals at C6 and C10, and log addresses for A7.

- Then decide whether to file requests with the City of Somerville or the state highway department, depending on who owns each crossing.
- B2 (the pole base) will be reported to the City once its exact address is logged.

What this report leaves out, by protocol

Health care buildings passed on the way are not named or described, and their entrances are not rated; building interiors were not entered. The entrance of a community organization's office is not rated either. Personal remarks in the narration and comments about people seen on the street are not used. Figures were chosen so that no faces or readable license plates appear. A search for an office at an address that turned out to be wrong is not reported as a missing sign.

Checking the AI-assisted analysis

The item list was drafted with AI help from the transcribed narration and the video frames. Several points needed correction or caution:

- One video was recorded with the phone sideways; its frames had to be rotated before they could be read.
- The exact places of C7 and C9 could not be read from the frames or the narration, so they are marked "to be confirmed" rather than guessed.
- Sizes (such as the 2-inch cracks in A7) and the short walk signal in C6 come from the narration and are estimates, not measurements.
- Items about care facilities and a community office entrance were removed by protocol, and a search at a wrong street number was dropped as a finding.
- Items with confidence M rest on the narration alone; they will be checked on the re-check walk (November 16 –28, 2026), and this report will be updated.

More photo evidence



Figure 7. The push button at the Broadway crossing on Winter Hill. Its spoken "wait" message was hard to hear over traffic (C1). The hand is the walker's.

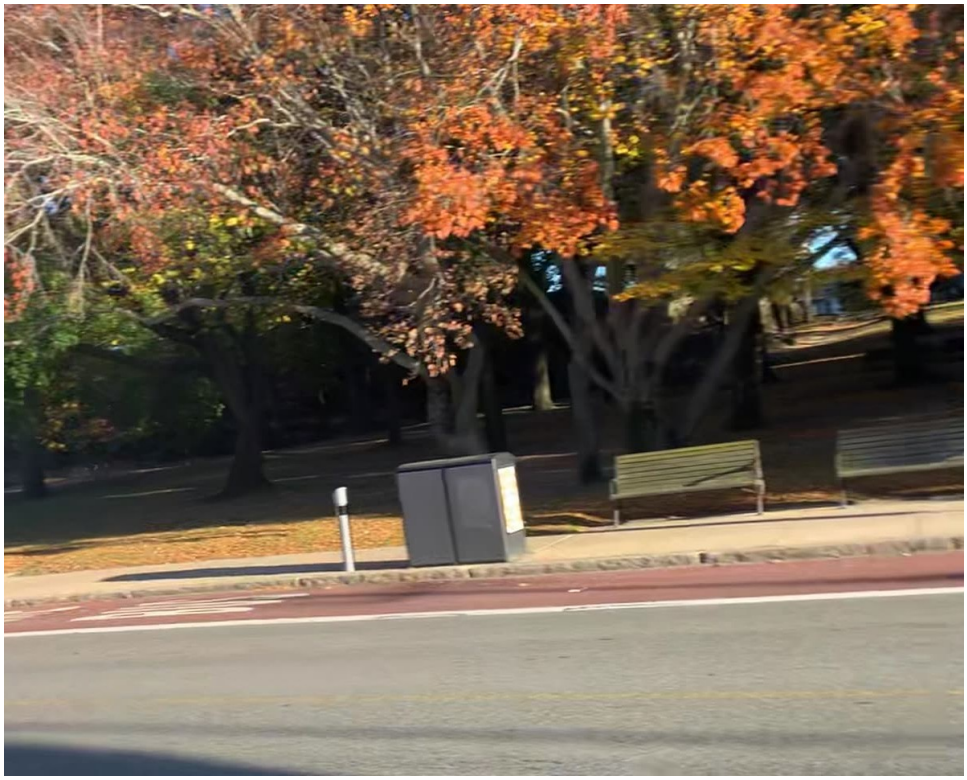


Figure 8. Powderhouse Square park: the first benches since Davis Square, about a mile into the walk (A5).



Figure 9. The crossing at Powderhouse Square: wide crosswalks, but nothing controlling traffic from one direction (A6).

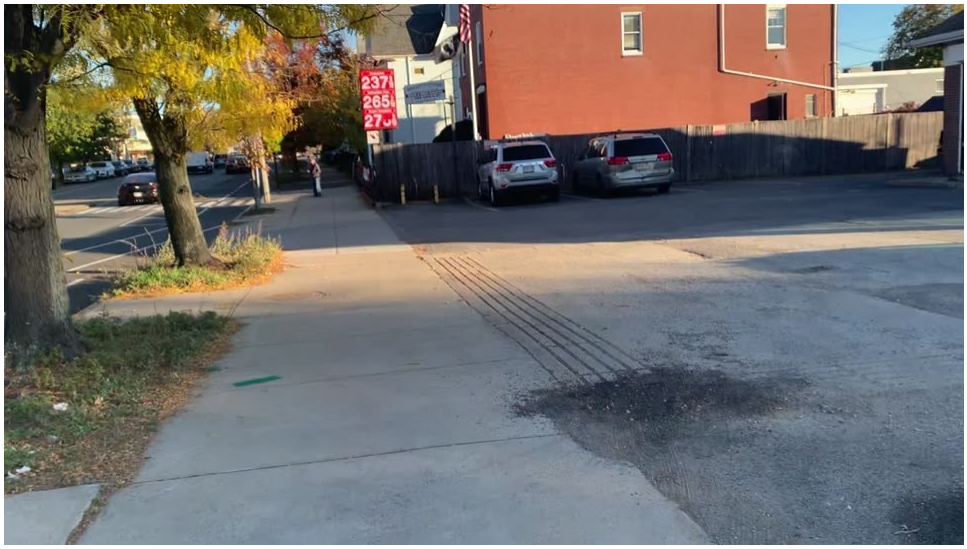


Figure 10. The driveway of a closed gas station on Broadway: a wide curb cut across the sidewalk with a broken edge (B4).

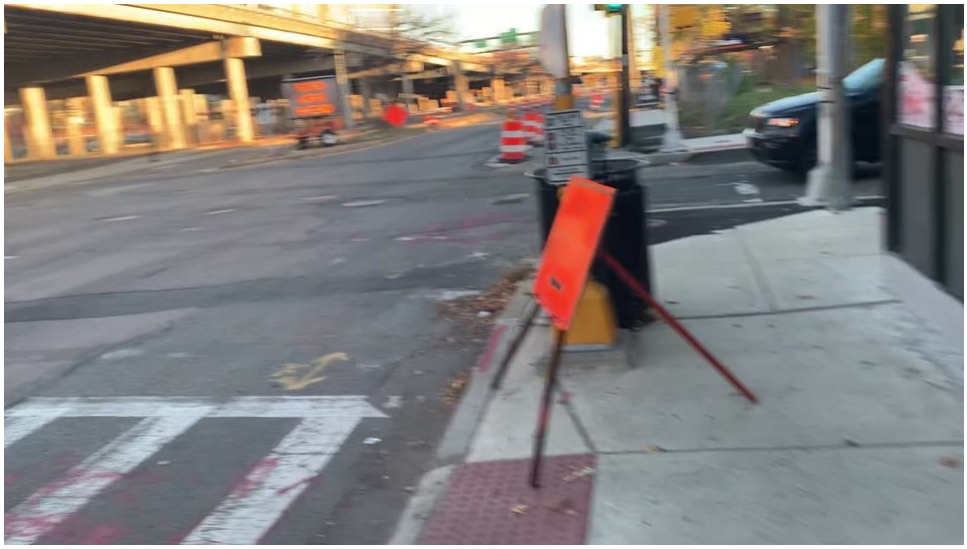


Figure 11. Mystic Ave under I-93: a detour sign standing on the sidewalk during construction (C6).



Figure 12. The store parking entrance at Assembly Square: broken asphalt right after the warning strip, and no crosswalk lines (C11).

How this report was produced

- **Capture:** three narrated smartphone videos made by the author.
- **Local processing on the author's computer:** audio extracted with ffmpeg (via imageio-ffmpeg); narration transcribed with openai-whisper, model "small"; one frame sampled every 5 seconds. Raw transcripts are not published.
- **AI review:** the frames and transcripts were reviewed by Claude (Anthropic), model claude-opus-5-5, which drafted the item list and this report. For this walk, frames were not blurred before the AI review; the figures in this report were chosen so that no faces or readable license plates appear. The author reviews the findings (see "Checking the AI-assisted analysis").
- **Limitations:** one walker on one dry afternoon; short stretches between the videos were not recorded; walk signals were not timed and slopes were not measured; the exact places of two high items are not yet confirmed; this was not a route to a research site; items are possible barriers, not legal compliance determinations. The items have not been reviewed by people with disabilities; feedback on the checklist is collected separately, so this is a preliminary assessment.

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